

ILOSTAT MCP — Use Case Specification

What this project is

Claude and other AI assistants are often asked labour market questions — unemployment rates, wage trends, country comparisons. They answer confidently even when they don't actually know. They produce plausible-sounding numbers from their training data, which may be outdated, approximated, or simply made up. There is no easy way to tell the difference.

The ILOSTAT MCP is a server that sits between Claude and ILOSTAT — the ILO's official labour statistics database — so Claude can look up real, verified numbers for 335 countries instead of guessing.

Two things make it more than a basic data lookup:

- 1. A hallucination benchmark** — a formal test measuring exactly how often Claude fabricates labour statistics versus how often it gets them right when equipped with this server. Scored across 30 questions and three conditions: no tools, full tools, tools without break detection. Not a demo — actual numbers.
- 2. Methodology break detection** — ILOSTAT's data sometimes switches survey source mid-series without any label (a country moves from a household survey to a labour force survey). Computing a trend across that switch produces a number that looks precise but is methodologically meaningless. This server detects those switches automatically and warns before any calculation. A bare AI assistant has no access to this information and will compute the trend anyway.

What you can ask

"Reliable" means: the same query always hits the same data source, applies the same filters, runs the same calculation, and returns the same number.

Indicator	Query type	Example question	What you get back
Unemployment	Single point	"What was Malaysia's unemployment rate in 2022?"	One number (%) for Malaysia 2022, with the survey source named
Unemployment	Single point	"What is France's current unemployment rate?"	Most recent available year's rate, with year stated explicitly
Unemployment	Single point	"Did Brazil's unemployment go up or down in 2018?"	The 2018 rate vs. 2017, with direction stated
Unemployment	Time series	"Show me Spain's unemployment rate from 2010 to 2023"	Year-by-year table. Break warning included if methodology changed mid-range.
Unemployment	Time series	"How has Thailand's unemployment changed	Year-by-year series for the last 10 available years. Clean series stated

Indicator	Query type	Example question	What you get back
		over the last decade?"	as such.
Unemployment	Time series	"What was Nigeria's unemployment each year between 2015 and 2022?"	Year-by-year series with explicit break warning at 2019 (survey source changed)
Unemployment	Year-on-year change	"What was the year-on-year change in Germany's unemployment from 2021 to 2022?"	Percentage point change between 2021 and 2022, with both values shown
Unemployment	Compound growth	"What's the CAGR of Pakistan's unemployment from 2010 to 2020?"	CAGR figure with explicit warning that 5 methodology breaks exist in this range
Unemployment	Trend	"Is Italy's unemployment trending up or down since 2015?"	Trend direction (up/down/flat), slope, and fit quality over the available data
Unemployment	Comparison	"Which has higher unemployment: Spain or Italy in 2022?"	Both values side by side, clear statement of which is higher
Unemployment	Comparison	"Compare unemployment trends in France and Germany from 2015 to 2023"	Both series side by side, trend direction for each, break warnings if applicable
Unemployment	Comparison	"Which improved more: Brazil or Argentina?"	Change for both countries over the period, with a clear comparative statement
Unemployment	Data existence	"Does ILOSTAT have unemployment data for Yemen?"	"No national survey data available for Yemen. ILO modelled estimates exist but are imputed, not observed."
Unemployment	Data existence	"What's the most recent unemployment data for Singapore?"	Latest available year and value
Wages	Single point	"What was the average monthly wage in Singapore in 2021?"	One number in local currency (SGD), year confirmed, survey source named
Wages	Single point	"What did workers earn on average in Germany in 2022?"	Average monthly wage in EUR for Germany 2022, survey source named
Wages	Single point	"What is France's average monthly wage?"	Most recent available year in EUR, year stated explicitly

Indicator	Query type	Example question	What you get back
Wages	Time series	"Show me Thailand's wage data from 2014 to 2022"	Year-by-year wages in THB, with break warnings at 2013 and 2014
Wages	Time series	"How have wages changed in Pakistan over the last decade?"	Year-by-year wage series with warnings for the multiple methodology breaks
Wages	Time series	"What's been happening to wages in Vietnam since 2018?"	Year-by-year series from 2018 to latest available, clean or with break warning
Wages	Year-on-year change	"How much did wages grow in France from 2019 to 2023?"	Percentage change 2019–2023, with both endpoint values shown
Wages	Compound growth	"What's the CAGR of wages in Germany from 2015 to 2022?"	CAGR figure, with break warning if a methodology change falls in that range
Wages	Trend	"Are wages in Singapore growing or declining?"	Trend direction, slope, fit quality over all available data
Wages	Comparison	"Who pays more: France or Germany?"	Both most-recent-year average monthly wages in local currency
Wages	Comparison	"Which had higher wage growth 2018–2022: Thailand or Vietnam?"	Change for both countries over the period, any break warnings, comparative statement
Wages	Comparison	"Compare wages in Singapore and Malaysia"	Both most-recent-year values in local currency, years stated
Wages	Data existence	"Does ILOSTAT have wage data for Georgia?"	"Yes — average monthly earnings data is available for Georgia (the country). If you meant the US state, ILOSTAT only covers countries."
Youth unemployment	Single point	"What was South Africa's youth unemployment in 2021?"	Youth unemployment rate for South Africa 2021, with note that "youth" here means ages 15–29 (the closest available ILOSTAT band; ILO's formal definition is 15–24)
Youth unemployment	Time series	"How has youth unemployment changed in South Africa over the past 5 years?"	Year-by-year youth unemployment series (ages 15–29), age band stated once

Packaged workflow

Workflow	How to trigger	What it does	What you get
Labour market snapshot	"Give me a labour market snapshot for Malaysia"	Chains: unemployment rate (latest year + year-on-year) → wage trend (last 5 years) → any break warnings for either series	Current unemployment rate and direction, wage trend and direction, any methodology caveats. All sourced from ILOSTAT survey data.

What you can't ask — and what the tool says

Question	Why it can't be answered	What the tool says
"What's unemployment for 25–34 year olds in France?"	ILOSTAT has no 25–34 age band. Available slices are 15–24 (youth) and 25+ (older adults).	"ILOSTAT doesn't publish unemployment data for the 25–34 age band. The closest available breakdowns are ages 15–24 or 25 and above. Would either of those help?"
"What's the male vs female wage gap in Germany?"	The tool retrieves male wages and female wages separately, but computing a gap is not a supported calculation.	"I can retrieve average monthly wages for German men and German women separately, but I don't compute a wage gap directly. Would you like both numbers so you can compare?"
"What are hourly wages in France?"	Hourly wages come from a different ILOSTAT dataset not covered in v1.	"This tool covers average monthly wages. ILOSTAT does publish hourly earnings separately, but that's not supported here in v1."
"What's unemployment by education level in Brazil?"	Education-disaggregated unemployment is a different dataset not covered in v1.	"ILOSTAT has unemployment broken down by education level, but that dataset isn't supported by this tool. I can give you Brazil's total unemployment rate."
"What's the labour force participation rate in Brazil?"	Labour force participation is a different indicator — not unemployment, not wages.	"Labour force participation isn't covered by this tool, which focuses on unemployment rates and average wages. ILOSTAT does publish that data — you'd need to query it directly."
"What's the rural unemployment rate in India?"	Urban/rural disaggregation is a different dataset with a different structure.	"Rural vs. urban unemployment breakdowns come from a different ILOSTAT dataset. I can give you India's national total unemployment rate."
"What industries have the highest wages in Germany?"	Wages by sector is a different dataset (earnings by economic activity).	"Wages broken down by industry aren't supported here. This tool covers average wages across all employees. ILOSTAT

Question	Why it can't be answered	What the tool says
		does publish sector-level earnings separately."
"What's France's GDP?"	GDP is a macroeconomic indicator from a different database (World Bank, IMF).	"GDP isn't in ILOSTAT — that's a World Bank or IMF indicator. This tool only covers employment and wage data."
"What will Malaysia's unemployment be in 2030?"	No observed data exists for 2030.	"There's no observed unemployment data for 2030 — it hasn't happened yet. I won't return ILO projections as if they were real survey data."
"What's the GCC unemployment rate?"	GCC is a regional bloc, not a country. ILOSTAT has no GCC aggregate code.	"ILOSTAT doesn't have a GCC aggregate — it's organised by country, not regional blocs. I can look up individual GCC members: Saudi Arabia, UAE, Qatar, Kuwait, Bahrain, or Oman."
"What's the World Bank's unemployment rate?"	The World Bank is an organisation, not a country or economy.	"The World Bank is an organisation — it doesn't have an unemployment rate. Did you mean a specific country?"
"What is the ILO's GDP?"	The ILO is an organisation; GDP is not a labour market indicator.	"The ILO is an international organisation — it doesn't have a GDP. And GDP is outside the scope of this tool regardless."
"What's the literacy rate in Malaysia?"	Literacy is a UNESCO/education indicator, not labour market data.	"Literacy rate is outside the scope of this tool. UNESCO and World Bank publish literacy statistics."
"What's the inflation rate in Germany?"	Inflation is a macroeconomic indicator, not a labour market indicator.	"Inflation data isn't in ILOSTAT. That's tracked by the IMF, World Bank, or Eurostat. This tool covers employment and wage data only."

Combo queries — how the MCP handles mixed and multi-part questions

Users rarely ask one clean question. The behaviour rule that applies to all combo queries:

Answer every in-scope part using tools. Explicitly decline every out-of-scope part and say why.
 Never answer an out-of-scope part from training knowledge — not even as a helpful supplement.
 Never silently skip part of a question; if you only partially answered it, say so.

The dangerous failure mode is not a wrong number — it is Claude answering the unsupported part from training memory and presenting it alongside tool results without flagging the source difference. That looks like a complete answer but mixes verified data with a hallucination.

Combo type	Example	How it is handled	What the user sees
In-scope + out-of-scope	"UK unemployment last 5 years and a 2-year forecast"	Tool call for the historical part returns clean data. Future years have no survey data — the tool returns <code>has_data: false</code> and does not use ILO projections as a substitute.	Historical series answered in full. Explicit statement that no survey-based forecast exists. Redirect to IMF World Economic Outlook for forecasts. No projection returned as if it were real data.
Two in-scope indicators	"What are unemployment AND wages in Germany in 2022?"	Two sequential tool calls — one for each indicator. Both return clean data.	Both numbers presented side by side, each with its source named and its unit stated.
In-scope + out-of-scope indicator	"Unemployment AND inflation for France"	One tool call for unemployment. No tool call for inflation — it is the wrong database.	Unemployment answered from ILOSTAT. Explicit statement that inflation is not in ILOSTAT and naming where to find it. Inflation figure NOT drawn from training knowledge.
In-scope with unavailable disaggregation	"Youth unemployment for ages 25–34 in Brazil"	Tool call returns youth unemployment for ages 15–29 — the closest available band. No 25–34 band exists.	Result returned with explicit statement of the actual age band used (15–29, not 25–34). User is told why the exact band they asked for is not available.
Ambiguous entity + valid query	"Unemployment in Georgia for the last 5 years"	No tool call made until ambiguity is resolved.	Claude asks: "Did you mean Georgia the country (in the Caucasus) or the US state of Georgia? ILOSTAT covers countries — if you mean the state, I don't have data for it." Tool call happens only after the user confirms.
Ambiguous entity — one option has data, one doesn't	"What is unemployment in Congo?"	No tool call until ambiguity is resolved. Republic of Congo (COG) returns 404; Democratic Republic of Congo (COD) has data. Claude cannot know this in advance — it must ask first.	Claude asks which Congo. After confirmation, tool call is made. If user says "Republic of Congo," tool returns <code>has_data: false</code> and Claude says so explicitly.

Combo type	Example	How it is handled	What the user sees
Multiple countries, one valid	"Compare unemployment in France and the GCC"	Tool call for France succeeds. GCC has no country code in ILOSTAT — not attempted.	France's unemployment returned. Explicit statement that GCC is not a country in ILOSTAT, with offer to look up individual member states instead.
In-scope calculation + unsupported calculation	"Male and female unemployment in Spain, plus the gender gap"	Two tool calls (male and female filters on the same flow). The gender gap is arithmetic Claude computes from those two numbers — not a tool call.	Male rate, female rate, and the computed gap all presented. Gap clearly labelled as derived from the two retrieved figures.
Fully out of scope	"GDP and inflation for Germany"	No tool calls. Nothing in this question is covered by the MCP.	Full decline: neither indicator is in ILOSTAT. Redirect to appropriate sources (World Bank for GDP, IMF/Eurostat for inflation). No figures drawn from training knowledge.